Do Plausible Market Simulators Recommend the Same Trade?

A Paired-Twin Study of Execution Decisions across ABIDES Configurations and a Retrained TRADES Market

Gaurav Chattree*

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Abstract

Market simulators are usually validated by how realistic their prices and order flow look. A trader uses them for something else: to decide how to execute. We ask whether simulators that all look plausible recommend the same execution schedule, and what it costs to trust one when another is true. One execution agent, one task and one cost accounting are run unchanged in six ABIDES configurations that differ only in their background population, and on a shorter task in a market driven by a TRADES diffusion model that we retrained on one public day of data. Costs are estimated against a same-seed twin with no intervention, across 7,680 ABIDES episodes. Four results follow. First, two of the six configurations, each one parameter away from the reference, are too unstable to evaluate anything (in one, the unperturbed price moves more than 1% within 30 minutes in 70% of runs), yet standard stylised facts do not flag them and they sit closest to a real reference day. Second, the four stable configurations agree: no policy pair is significantly ranked in opposite order in two of them, all four select an even schedule for an order of 20% of volume, and the largest regret from choosing a schedule in one and deploying it in another is 1.5 basis points. Third, a mean-field-game optimiser calibrated in each world does prescribe different schedules in different worlds, but the differences are calibration noise, and its front-loaded prescription costs two to three times as much as an even schedule where the difference is measurable. The retrained TRADES market, by contrast, gives the opposite advice to ABIDES (sell at once), because its quoted spread widens steadily once generation takes over from replay; this disagreement is real but is an artefact that a spread-versus-time check exposes. We did not detect a relation between the distance of two worlds' stylised facts and the regret of transferring between them, at a resolution too coarse to rule one out. ABIDES-MARL could not be included because no public implementation was found, and all regret is simulator-to-simulator: nothing here is validated against a real market.

1 Introduction

A trader who must sell a block of shares before a deadline faces a familiar trade-off: selling quickly moves the price against them, selling slowly leaves them exposed to the market for longer. Because a real market cannot be re-run, the schedule is almost always chosen with the help of a *market simulator*: either a model with explicit impact parameters (Almgren and Chriss, 2001), an agent-based simulation such as ABIDES (Byrd et al., 2019; Amrouni et al., 2021), or, more recently, a learned generator of order flow (Coletta et al., 2021; Berti et al., 2025; Zhang et al., 2026).

*Experiments, code and manuscript were produced with the assistance of an AI coding agent (Claude Code); all numbers are generated by the scripts in the accompanying repository. Co-authors to be added.

Simulators are usually judged by *observational realism*: whether spreads, volatility, order-flow statistics and other stylised facts look like the real thing (Vyetrenko et al., 2020; Nagy et al., 2025). A trader, however, does not consume stylised facts. They consume a *decision*: which schedule to run. This paper asks the decision-level question directly.

> If several simulated markets are all plausible, do they recommend the same execution schedule, and what does it cost to trust one of them when another one is true?

We answer it with a controlled experiment. One execution agent, one task definition and one cost accounting are run unchanged in 6 ABIDES market configurations that differ only in their background population, and, on a shorter task, in a market driven by the TRADES diffusion model (Berti et al., 2025). Costs are estimated with *paired twins*: every intervention episode is compared with an episode that has the same seed and no intervention, so that background noise cancels. In total the study uses 7,680 simulated ABIDES episodes and 36 TRADES episodes.

The decision-maker we study is a mean-field-game (MFG) liquidator in the tradition of Cardaliaguet and Lehalle (2018), Huang et al. (2019) and Casgrain and Jaimungal (2020). In its risk-neutral, homogeneous form the MFG equilibrium is a one-parameter family of schedules, which lets us define "the schedule that simulator i recommends" in two independent ways: by calibrating the MFG's impact parameters from probe trades in simulator i, and by direct search over the same family inside simulator i.

Contributions.

1. A reproducible harness that injects an identical execution agent into ABIDES configurations and into a TRADES-driven market, with paired-twin cost estimation (Section 3).
2. A decision-level comparison: policy rankings, the schedule each world recommends, and the full train-in-i/deploy-in-j regret matrix (Sections 4.2–4.3).
3. A test of whether differences in standard stylised facts between two worlds predict the regret of transferring a schedule between them (Section 4.4).
4. An assessment of an MFG optimiser calibrated in each world, including where its structural assumptions fail (Section 4.5).

Scope, stated up front. The original design called for three systems: ABIDES, ABIDES-MARL (Cheridito et al., 2025) and TRADES. We could not find a public implementation of ABIDES-MARL, so it is absent. The authors' TRADES checkpoints are no longer downloadable; we retrained the TRADES architecture ourselves on the single public LOBSTER sample day, for under two hours on a laptop, so the TRADES arm tests *our retrained model*, not the published one. All regret figures are simulator-to-simulator; nothing here measures regret against a real market. Section 6 lists every deviation.

2 Related work

Agent-based market simulation. ABIDES (Byrd et al., 2019) simulates a NASDAQ-style exchange with message-passing agents; its behaviour is determined by the configured agent population, not by training data. ABIDES-Gym (Amrouni et al., 2021) exposes it as a reinforcement-learning environment, and ABIDES-MARL (Cheridito et al., 2025) extends that to several simultaneously

learning agents and studies an execution problem with strategic liquidity providers. Balch et al. (2019) argue that interactive simulation, unlike market replay, lets a strategy's own impact feed back into prices; that feedback is exactly what our comparison probes.

Learned order-flow generators. Coletta et al. (2021) condition a GAN on the order-book state to generate reactive order flow. TRADES (Berti et al., 2025) replaces the GAN with a transformer-based diffusion model and deploys it as a "world agent" inside an ABIDES exchange; its authors evaluate realism, responsiveness to an injected agent, and usefulness of the synthetic data, and note that strategy evaluation needs further work. M3 (Zhang et al., 2026) is a recent token-based foundation model for order flow with meta-order impact experiments. Our contribution is not another generator but a decision-level comparison across simulators.

Evaluating simulators. Realism metrics for limit-order-book simulators are surveyed and standardised by Vyetrenko et al. (2020) and by LOB-Bench (Nagy et al., 2025), which also includes price-response statistics. These evaluate what a simulator looks like. We ask the complementary question of whether simulators that look alike also *recommend* alike.

Optimal execution and mean-field games. Almgren and Chriss (2001) derive the classical single-agent schedule under linear temporary and permanent impact. Mean-field-game formulations replace the single agent with a population whose aggregate flow moves the price (Cardaliaguet and Lehalle, 2018; Huang et al., 2019; Casgrain and Jaimungal, 2020). Empirically, the impact of a meta-order is concave in its size and partly reverts after it ends (Tóth et al., 2011); both features matter for the results below.

3 Experimental setup

3.1 Worlds

A *world* is a complete background market. Table 1 lists the 6 ABIDES worlds. All share the ABIDES-Markets exchange (price–time priority, one-cent tick), the RMSC04 reference parameters, and the same fundamental-value oracle; they differ in one population parameter each. World T1 is the TRADES market described in Section 3.5. Because every world shares an ABIDES-family matching engine, the study compares *background-flow models*, not independent market infrastructures.

Table 1: ABIDES worlds. Volume is mean background shares traded 10:00–10:30 over unperturbed runs.

World	Background population (change from RMSC04)	Volume
A1	RMSC04 reference: 1000 noise, 102 value, 12 momentum traders, 2 market makers	81,826
A2	no market makers	74,370
A3	36 momentum traders (3×)	167,719
A4	40 value traders (weaker anchoring, less flow)	50,887
A5	4 market makers at 4× participation	112,367
A6	4× fundamental volatility	81,669

3.2 Task, policies and execution agent

The task is to sell (or, for the asymmetry check, buy) Q shares between 10:00 and 10:30 using market orders at $n = 60$ decision times 30 seconds apart. Q is set per world to 1%, 5% and 20% of that world's own mean background volume over the window, measured from unperturbed runs, so "5%" means the same participation everywhere. The market opens at 09:30; the first 30 minutes are warm-up. The simulation continues for five minutes after the deadline to observe reversion.

One execution agent implements every policy. At each decision time it queries the book and recent volume, asks the policy for a child size, and sends a single market order; the last slice always sends the remainder. Policies are:

- **Schedule family.** $w_k \propto \exp(-b\,k/n)$ for $b \in \{-2, 0, 2, 4, 8\}$: $b = 0$ is TWAP, $b > 0$ front-loads, $b < 0$ back-loads.
- **Immediate.** The whole parent order as one market order at 10:00.
- **POV.** A fixed fraction of the market volume observed over the previous slice.
- **Urgency rule.** Deadline-paced, trading twice as fast when the spread is at most two ticks and half as fast when it is six ticks or more.

We did not train a reinforcement-learning policy; the planned RL arm was replaced by exhaustive search over the schedule family, which gives a deterministic "optimised in world i" policy without optimiser noise.

3.3 Paired-twin cost estimation

For world w and seed s we run a *twin* episode in which the execution agent is present, queries the market at the same times, and trades nothing. All background random streams are seeded before the execution agent is created, so twin and intervention are identical until the first child order. We verified, for every world, that the same seed reproduces the same mid-price path bit-for-bit (also across operating-system processes), that the pre-10:00 paths of twin and intervention are identical, and that they differ afterwards.

Let q_k be the shares filled in slice k at average price p_k, let m_k^0 be the twin's mid price at that decision time, and let m_0^0 be the arrival mid. The *paired impact cost* of a sell is

$$C \;=\; \frac{10^4}{m_0^0}\,\frac{\sum_k q_k\,(m_k^0 - p_k)}{\sum_k q_k} \quad \text{(basis points)}. \tag{1}$$

It equals implementation shortfall minus the part of the shortfall that the market would have produced without us, so its variance across seeds is far smaller than that of raw shortfall. Raw shortfall is reported alongside. If the book cannot absorb an order, the unfilled shares are charged at the worst per-share cost of any slice in that episode; fill fractions are reported separately. The mid-price *displacement* is the adverse difference between intervention and twin mid prices.

3.4 Mean-field-game schedule

Consider a population of identical sellers, each holding normalised inventory q_t with $q_0 = 1$ and selling at rate $\nu_t = -\dot{q}_t$ on $t \in [0, 1]$. An agent receives $S_t - a\,\nu_t$ per share, where a is a temporary-impact coefficient, while the mid price is pushed by the population's mean rate $\bar{\nu}_t$: $\mathrm{d}S_t = -\lambda\,\bar{\nu}_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$. With a running inventory penalty ϕ, an agent that takes $\bar{\nu}$ as given minimises $\mathbb{E}\int_0^1 (a\nu_t^2 +$

$\phi q_t^2 + \lambda \bar{\nu}_t q_t)\,\mathrm{d}t$ subject to $q_1 = 0$. The Euler–Lagrange condition is $2a\ddot{q} = 2\phi q + \lambda\bar{\nu}$, and imposing the equilibrium condition $\bar{\nu} = \nu = -\dot{q}$ gives

$$\ddot{q} + \frac{\lambda}{2a}\,\dot{q} - \frac{\phi}{a}\,q = 0, \qquad q_0 = 1,\ q_1 = 0. \tag{2}$$

For a risk-neutral population ($\phi = 0$) the solution is $\nu_t \propto e^{-bt}$ with

$$b_{\mathrm{MFG}} = \frac{\lambda}{2a}, \tag{3}$$

exactly our schedule family. Two reference points follow. A single risk-neutral Almgren–Chriss trader, who internalises their own permanent impact, is indifferent to λ and trades TWAP ($b = 0$). The MFG population front-loads because each member races the others' impact. We inject the equilibrium aggregate flow as one meta-order.

Calibration in world w uses only that world's training seeds: a is the slope of per-slice slippage (fill price against the agent's own pre-trade mid) on the normalised child size, pooled over the schedule-family runs; λ is the mean mid displacement at the deadline under TWAP. The cost of an off-grid b is obtained by linear interpolation of the estimated cost curve, and b_{MFG} is clipped to the grid range.

3.5 TRADES arm

TRADES (Berti et al., 2025) is released as the DeepMarket repository, which embeds a "world agent" in a legacy ABIDES exchange: it replays the first 15 minutes of a real LOBSTER day (Huang and Polak, 2011) and then generates each subsequent order with a diffusion model conditioned on the previous 255 orders and the live order book. The authors' checkpoints are hosted in a cloud folder that has been deleted (an open issue on the repository reports the same), and their training data is licensed. We therefore retrained the architecture, with the repository's fixed hyperparameters (10.0 M parameters), on the one public LOBSTER sample day bundled with the repository (INTC, 21 June 2012, 520,584 training events after an 85/5/10 chronological split of the day). Training ran for 812 gradient steps at batch size 256 on a laptop GPU, about 40% of one pass over the data; validation loss fell from 3.03 at the first check to 2.33 at the last. This is a small fraction of the authors' training budget, and the model saw one stock on one day.

We added an execution agent with the same semantics as in ABIDES and restored the aggressor's fill notification, which the repository disables. Because a TRADES episode costs minutes rather than seconds, this arm uses a shorter task: sell 10% of background volume between 09:46 and 09:56 in 20 slices, with 6 seeds and four schedules plus the immediate order. The same 10-minute task is run in all ABIDES worlds so the two families are compared on identical terms.

3.6 Compute and reproducibility

Everything ran on one Apple M1 laptop (16 GB). The code pins ABIDES (`abides-jpmc-public` commit `f9cbe51`) and DeepMarket (commit `8f1f89b`) and ships the patch applied to the latter. Seeds are split in half: schedules are selected on the first half and all transfer costs and regrets are reported on the second.

4 Results

The main experiment ran 7,680 ABIDES episodes: 200 unperturbed twins per world and, for each of 40 seeds, eight sell policies and one buy policy at three sizes in six worlds. In total, 1 unperturbed

and 5 intervention episodes hit the wall-clock limit on every attempt (A3 seed 1, A3 seed 118, A3 seed 16); they are excluded together with all other episodes of the same seed, leaving 38 of 40 seeds in the main analysis. Selection of schedules uses the first 19 seeds and transfer results the remaining 19.

4.1 Two of the six worlds are not stable enough to evaluate anything

Table 2 gives each world's stylised facts with no intervention. All six have tight spreads, heavy-tailed returns and volatility clustering, and none would be rejected on those grounds. The last two simulated rows tell a different story. In A3 (three times the momentum traders) the mid price moves by more than 1% within the 30-minute window in 70% of unperturbed runs, with a standard deviation of 488 bps; in A4 (fewer value traders) the same happens in 13% of runs. A1, A2, A5 and A6 never do. We call the latter four *stable* and report decision results for them separately; the rule (no unperturbed run moves by more than 1%) was fixed after seeing these numbers, so it is descriptive, not pre-registered.

Table 2: Stylised facts of each unperturbed world over the task window (means over 200 seeds; 199 in A3). INTC is the real LOBSTER sample day (21 June 2012, 10:00–10:30), a single reference point. Distance to INTC: root-mean-square difference over spread, volatility, the two return autocorrelations, kurtosis and order-flow autocorrelation, each scaled by its standard deviation across worlds.

Statistic	A1	A2	A3	A4	A5	A6	INTC
Quoted spread (bps)	0.17	0.38	0.90	0.67	0.14	0.31	3.66
Depth, top 5 levels (shares)	696	664	675	452	1593	675	63636
10 s return volatility (bps)	0.37	0.83	5.69	2.17	0.21	0.94	3.08
Return autocorrelation, lag 1	0.06	-0.12	-0.01	0.00	0.01	0.15	0.04
|return| autocorrelation, lag 1	0.22	0.25	0.36	0.26	0.09	0.31	0.04
Excess kurtosis	9.6	16.7	18.8	19.1	3.2	15.7	14.3
Mean trade size (shares)	55	58	61	52	71	54	295
Order-flow imbalance autocorr.	0.25	0.20	0.10	0.08	0.35	0.25	-0.21
30-min mid move, s.d. (bps)	9.6	10.1	488.3	200.8	9.0	31.1	–
Share of runs with |move| $> 1\%$	0.00	0.00	0.70	0.13	0.00	0.00	–
Distance to INTC (scale-free statistics)	5.71	5.40	4.67	4.79	5.89	5.61	0

The instability is not visible in the conventional statistics. Measured against the one real reference day, on the six scale-free statistics, the two unstable worlds are the *closest* to the real market (last row of Table 2), largely because their volatility and spread are nearer to INTC's than those of the stable worlds, whose \$1000 price and one-cent tick make spreads unrealistically tight. With one real day this ordering is anecdotal, but it is a concrete case in which a distance on standard stylised facts favours simulators that cannot support an execution experiment.

Instability also defeats the paired estimator (Table 3). In the stable worlds, pairing with a twin cuts the standard deviation of the cost estimate by a factor of roughly two to four. In A3 it removes nothing: once the first child order perturbs the market, twin and intervention paths separate so quickly that their difference is as noisy as either path. Every A3 and A4 number below should be read with that in mind.

Table 3: Standard deviation across seeds (bps) of raw implementation shortfall and of the paired impact cost, TWAP sell. Pairing removes most of the noise in the stable worlds and little of it where paths diverge quickly.

Size	Estimator	A1	A2	A3	A4	A5	A6
1%	raw shortfall	4.9	5.5	131.4	46.7	5.7	18.5
	paired cost	1.9	2.1	112.6	34.4	1.5	7.0
5%	raw shortfall	5.2	5.0	208.4	30.2	5.6	18.5
	paired cost	2.5	2.5	208.9	14.1	1.7	8.0
20%	raw shortfall	5.0	5.7	5499.9	230.8	6.1	16.5
	paired cost	2.4	3.0	5472.4	225.8	2.4	9.5

4.2 Policy costs and rankings

Table 4 reports the paired impact cost of every policy, and Table 5 the paired difference of each policy from TWAP.

Table 4: Paired impact cost of selling, in basis points (mean ± half-width of the 95% bootstrap interval over seeds). Bold marks the lowest mean in each column; many differences within a column are inside the intervals.

Policy	A1	A2	A3	A4	A5	A6
Parent order = 1% of background volume						
Schedule $b = -2$ (back-loaded)	**-0.0 ±0.8**	0.5 ±0.8	**-19.3 ±58.7**	-0.1 ±4.7	**-0.3 ±0.6**	**-0.0 ±2.4**
Schedule $b = 0$ (TWAP)	0.3 ±0.6	0.3 ±0.7	2.6 ±33.3	4.1 ±10.6	0.2 ±0.5	0.2 ±2.2
Schedule $b = 2$	0.3 ±0.4	0.2 ±0.5	-1.8 ±20.9	1.9 ±4.2	-0.0 ±0.3	1.0 ±1.4
Schedule $b = 4$	0.3 ±0.2	0.1 ±0.4	-13.3 ±25.5	-0.4 ±3.4	0.2 ±0.2	0.2 ±0.8
Schedule $b = 8$ (front-loaded)	0.2 ±0.1	0.2 ±0.2	-2.2 ±8.5	0.2 ±0.6	0.2 ±0.1	0.6 ±0.5
Immediate	0.8 ±0.3	0.8 ±0.3	2.0 ±0.4	1.2 ±0.5	0.4 ±0.1	1.3 ±0.5
POV	0.1 ±0.8	0.6 ±0.7	33.7 ±52.6	16.9 ±18.2	0.0 ±0.6	1.3 ±2.0
Urgency rule	0.3 ±0.4	**-0.2 ±0.6**	50.1 ±89.6	**-4.6 ±5.9**	0.1 ±0.3	0.7 ±2.0
Parent order = 5% of background volume						
Schedule $b = -2$ (back-loaded)	1.4 ±0.9	**0.2 ±1.1**	**-45.4 ±129.5**	1.0 ±11.9	0.7 ±0.7	3.4 ±3.0
Schedule $b = 0$ (TWAP)	0.8 ±0.8	0.4 ±0.8	-16.8 ±66.1	4.5 ±4.4	0.6 ±0.5	2.1 ±2.5
Schedule $b = 2$	0.8 ±0.5	0.2 ±0.5	-3.8 ±31.7	0.7 ±10.0	0.5 ±0.3	1.0 ±1.4
Schedule $b = 4$	0.8 ±0.3	0.5 ±0.4	20.6 ±28.1	**0.3 ±8.9**	0.6 ±0.2	**0.8 ±1.2**
Schedule $b = 8$ (front-loaded)	0.9 ±0.2	0.8 ±0.2	17.7 ±33.8	0.4 ±3.4	0.8 ±0.1	1.1 ±1.0
Immediate	2.5 ±0.3	2.1 ±0.4	6.5 ±3.4	4.3 ±2.3	1.9 ±0.5	5.0 ±1.8
POV	1.4 ±0.7	0.6 ±0.7	62.0 ±83.9	4.6 ±4.1	0.6 ±0.5	1.4 ±2.7
Urgency rule	**0.8 ±0.5**	0.3 ±0.7	-17.1 ±40.2	4.0 ±8.4	**0.4 ±0.3**	0.9 ±2.0
Parent order = 20% of background volume						
Schedule $b = -2$ (back-loaded)	3.5 ±1.3	2.7 ±1.3	33.8 ±305.1	108.3 ±99.6	2.0 ±0.8	5.2 ±5.1
Schedule $b = 0$ (TWAP)	**1.9 ±0.8**	1.6 ±0.9	-1051.6 ±1577.8	26.2 ±74.3	**1.5 ±0.7**	**4.0 ±3.0**
Schedule $b = 2$	2.4 ±0.6	**1.5 ±0.6**	0.4 ±164.4	60.8 ±66.9	2.1 ±0.5	4.8 ±2.1
Schedule $b = 4$	3.3 ±1.3	1.9 ±0.6	47.3 ±208.5	**-560.5 ±976.3**	2.4 ±0.4	4.2 ±1.4
Schedule $b = 8$ (front-loaded)	5.3 ±2.5	2.5 ±0.6	-36.9 ±91.7	1.3 ±36.7	3.2 ±0.6	5.5 ±1.4
Immediate	5.8 ±1.2	2.6 ±0.6	12.9 ±7.8	12.6 ±9.6	3.1 ±0.4	8.9 ±2.2
POV	2.4 ±0.6	1.6 ±0.8	**-1272.2 ±2012.7**	65.0 ±152.6	2.0 ±0.5	4.6 ±1.9
Urgency rule	2.1 ±0.7	1.5 ±0.7	244.9 ±269.2	30.0 ±74.6	2.4 ±0.6	4.5 ±2.7

Three things stand out in the stable worlds.

At 1% of volume nothing is resolvable. All costs are below about 1.5 bps and no policy differs from TWAP by more than its interval, apart from one isolated cell. Table 5 contains 126

Table 5: Paired difference in impact cost between each policy and TWAP, in bps (positive: worse than TWAP), over all seeds. Bold: the 95% bootstrap interval of the paired difference excludes zero.

Policy minus TWAP	A1	A2	A3	A4	A5	A6
Parent order = 1% of background volume						
Schedule $b = -2$ (back-loaded)	-0.3	+0.2	-22.0	-4.2	**-0.5**	-0.3
Schedule $b = 2$	-0.0	-0.2	-4.4	-2.2	-0.2	+0.8
Schedule $b = 4$	+0.0	-0.3	-15.9	-4.5	-0.0	-0.0
Schedule $b = 8$ (front-loaded)	-0.1	-0.2	-4.8	-3.8	+0.0	+0.4
Immediate	+0.5	+0.4	-0.6	-2.9	+0.2	+1.1
POV	-0.2	+0.3	+31.1	+12.9	-0.2	+1.1
Urgency rule	+0.0	-0.6	+47.4	-8.7	-0.1	+0.4
Parent order = 5% of background volume						
Schedule $b = -2$ (back-loaded)	+0.5	-0.2	-28.6	-3.5	+0.0	+1.2
Schedule $b = 2$	-0.0	-0.2	+13.0	-3.8	-0.1	-1.2
Schedule $b = 4$	-0.0	+0.1	+37.4	-4.2	-0.0	-1.4
Schedule $b = 8$ (front-loaded)	+0.1	+0.4	+34.5	-4.1	+0.1	-1.0
Immediate	**+1.7**	**+1.8**	+23.3	-0.2	**+1.3**	+2.8
POV	+0.6	+0.2	+78.8	+0.1	-0.0	-0.8
Urgency rule	-0.0	-0.1	-0.3	-0.5	-0.2	-1.3
Parent order = 20% of background volume						
Schedule $b = -2$ (back-loaded)	**+1.6**	+1.1	+1085.5	+82.1	+0.5	+1.2
Schedule $b = 2$	+0.5	-0.2	+1052.0	+34.6	+0.6	+0.8
Schedule $b = 4$	**+1.4**	+0.3	+1099.0	-586.7	**+0.9**	+0.2
Schedule $b = 8$ (front-loaded)	**+3.5**	+0.8	+1014.7	-24.9	**+1.7**	+1.5
Immediate	**+3.9**	+1.0	+1064.6	-13.5	**+1.6**	**+4.9**
POV	+0.5	+0.0	-220.5	+38.8	+0.5	+0.6
Urgency rule	+0.2	-0.2	**+1296.5**	+3.8	**+0.9**	+0.5

comparisons with no multiplicity correction, so a handful of bold cells are expected by chance.

At 5%, only the immediate order is distinguishable. It costs 1.3 to 1.8 bps more than TWAP in A1, A2 and A5, with intervals excluding zero. The five schedules, POV and the urgency rule are statistically tied.

At 20%, the worlds agree on direction and differ in strength. Front-loading is penalised: $b = 8$ costs 3.5 bps more than TWAP in A1 and 1.7 bps more in A5, both significant, and 0.8 and 1.5 bps more in A2 and A6, not significant. Back-loading is also penalised in A1 (+1.6 bps). The cost curve is U-shaped around TWAP (Figure 1). So the reference world and the deep-market-maker world punish urgency clearly, the world without market makers barely does, and none of them rewards it.

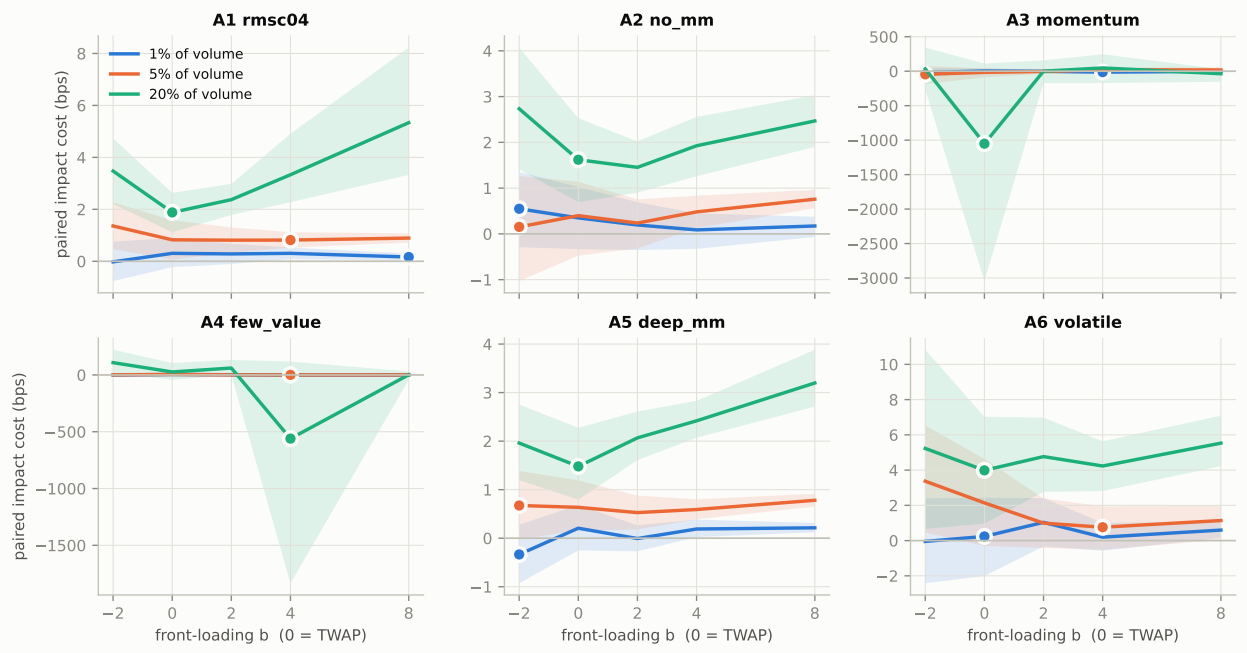


Figure 1: Paired impact cost of the schedule family in each world. Bands are 95% bootstrap intervals over 38 seeds; the ringed marker is the schedule selected on training seeds. Vertical scales differ by two orders of magnitude between stable worlds (A1, A2, A5, A6) and unstable ones (A3, A4).

Table 6 summarises agreement between rankings. Across the stable worlds the mean Spearman correlation of the eight policies' costs is 0.62 at 5% and 0.61 at 20%, and there is not one significant reversal: no pair of policies is significantly ordered one way in one stable world and the other way in another. Including the unstable worlds lowers the correlations and produces six nominal reversals at 20%, all involving A3 or A4.

4.3 Which schedule each world recommends, and the cost of trusting the wrong world

Selecting the cheapest schedule on training seeds gives each world's recommendation $b_i^\star$; Figure 2 shows the regret $R_{ij} = C_j(b_i^\star) - C_j(b_j^\star)$ on test seeds, and Appendix B gives the numbers.

At 20% all four stable worlds, and A3, select TWAP, so regret among them is exactly zero. At

Table 6: Agreement of policy rankings between worlds, over world pairs. ρ: Spearman correlation of the eight policies' mean costs. A policy pair is *flipped* if its order differs between two worlds, and a *significant reversal* if both paired 95% intervals exclude zero with opposite signs.

Size	Mean ρ	Min ρ	Max ρ	Mean flipped pairs	Significant reversals
All six worlds					
1%	0.25	-0.31	0.67	11.1 / 28	0 / 420
5%	0.39	-0.24	0.83	10.0 / 28	0 / 420
20%	0.20	-0.69	0.83	11.9 / 28	6 / 420
Stable worlds (A1, A2, A5, A6)					
1%	0.25	-0.21	0.62	11.7 / 28	0 / 168
5%	0.62	0.19	0.83	7.2 / 28	0 / 168
20%	0.61	0.24	0.83	7.8 / 28	0 / 168

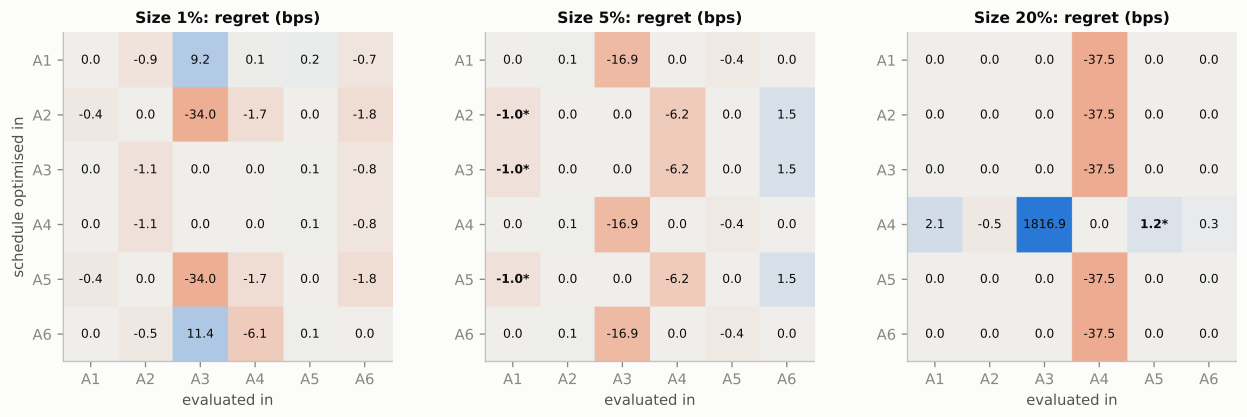


Figure 2: Transfer regret in bps (test seeds). Rows: world in which the schedule was selected. Columns: world in which it is evaluated. Bold with an asterisk: paired 95% interval excludes zero. Colour is on a signed logarithmic scale. Negative entries arise when the schedule selected on a world's own training seeds turns out worse on its test seeds than another world's choice, i.e. they measure selection noise.

1% and 5% the selected b scatters across the grid (for example -2 in A2 and A5, 4 in A1 and A6 at 5%), but the scatter is selection noise on flat cost curves: among stable worlds the largest regret is 1.5 bps, the mean is -0.2 bps, and 0 of 36 off-diagonal regrets is significantly positive.

The only significantly positive regret in the whole matrix has an unstable world as its source: at 20%, A4's noisy cost curve selects $b = 4$, which costs 1.2 bps more than TWAP in A5 (significant) and 2.1 bps more in A1 (not significant). That is the practical risk in this study: not that plausible stable simulators disagree, but that an unstable one returns a confident-looking recommendation that is noise.

4.4 Does observational distance predict transfer regret?

For each ordered pair of worlds we compare the distance between their stylised facts (z-scored across worlds) with the regret of transferring a schedule between them. Pooled over sizes, the Spearman correlation is 0.08 over 90 size-pair observations averaged to 30 pairs (permutation $p = 0.55$, permuting world labels); by size it is 0.23, -0.29 and 0.12. Restricted to the 12 pairs of stable worlds it is 0.25. We find no evidence that observational distance predicts regret (Figure 3).

This is a weak test. Regret among stable worlds is at the noise floor, so there is almost nothing to predict, and six configurations of one simulator family give at most 30 dependent pairs. The result should be read as "not detected at this resolution", not as evidence of independence.

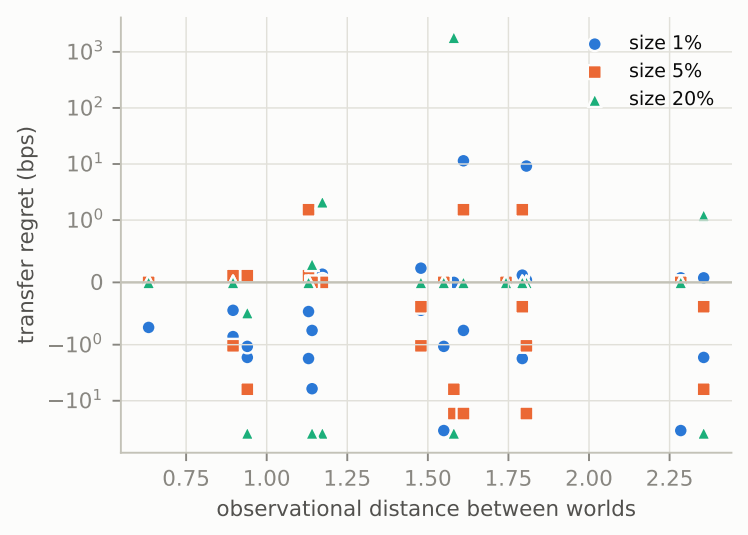


Figure 3: Observational distance between two worlds against the regret of transferring a schedule from one to the other. The vertical axis is linear within ± 1 bp and logarithmic beyond.

4.5 The MFG optimiser

Table 7 shows the MFG calibration in each world and the schedule it prescribes through Equation (3).

The prescription is unstable across worlds, and the instability is calibration noise. Temporary impact at the TWAP rate is tiny (a few hundredths of a basis point at 1% and 5%, about half a basis point at 20%), while the displacement λ is a few basis points with an interval that includes zero in most stable cells. Their ratio therefore swings to the edge of the grid: b_{MFG} is clipped in 15 of 18 cells. At 20% it is 3.0 in A1, -1.4 in A2, 8 in A5 and 4.2 in A6, while direct search selects

Table 7: MFG calibration per world (training seeds) and the schedule it prescribes, against the schedule chosen by direct search ($b^\star_{\text{sim}}$, training seeds). a: temporary impact at the TWAP rate (bps); λ: mean mid displacement at the deadline under TWAP (bps). Costs are test-seed means in bps. * clipped to the grid range $[-2, 8]$.

World	a	λ [95% CI]	b_{MFG}	$b^\star_{\text{sim}}$	$C(b_{\text{MFG}})$	$C(\text{TWAP})$	$C(b^\star)$
Parent order = 1% of background volume							
A1	0.01	1.2 [-1.0, 3.5]	8.0*	8	0.1	0.1	-0.3
A2	0.02	0.3 [-1.7, 2.3]	8.0*	-2	0.1	0.5	-0.1
A3	-0.43	-24.5 [-159.9, 131.4]	8.0*	4	-8.1	-5.9	-51.3
A4	-0.04	33.0 [5.5, 69.4]	8.0*	4	0.3	-6.0	-6.0
A5	0.00	0.5 [-1.7, 2.6]	8.0*	-2	0.2	0.1	-0.0
A6	0.01	-3.6 [-13.0, 5.1]	-2.0*	0	-0.0	1.8	-0.0
Parent order = 5% of background volume							
A1	0.05	4.6 [1.9, 7.3]	8.0*	4	0.8	0.0	-0.3
A2	0.08	1.4 [-0.5, 3.5]	8.0*	-2	0.8	0.0	0.0
A3	0.22	-12.3 [-117.6, 63.7]	-2.0*	-2	2.8	-68.3	-68.3
A4	-0.01	28.4 [12.1, 47.7]	8.0*	4	-0.3	-0.9	-8.4
A5	0.03	2.4 [0.0, 4.6]	8.0*	-2	0.8	0.3	0.3
A6	0.08	3.8 [-4.8, 11.7]	8.0*	4	1.5	2.9	0.5
Parent order = 20% of background volume							
A1	0.51	3.1 [1.0, 5.0]	3.0	0	2.9	1.5	1.5
A2	0.37	-1.1 [-3.3, 1.2]	-1.4	0	2.4	2.4	1.6
A3	-2.13	-1688.9 [-5207.2, 544.2]	8.0*	0	-100.1	-1812.1	-1812.1
A4	-0.18	558.9 [14.6, 1556.5]	8.0*	4	-22.0	-25.2	-25.2
A5	0.14	3.0 [-0.2, 6.3]	8.0*	0	2.9	1.0	1.0
A6	0.69	5.8 [-2.5, 15.2]	4.2	0	4.8	4.4	0.7

0 in all four. So the answer to "does the MFG optimiser decide differently in different worlds?" is yes, but for a reason that has little to do with the worlds: the decision rule divides a noisy number by a small one.

Where it can be resolved, the MFG prescription is costly for a single meta-order. In A1 and A5 at 20%, front-loading to $b = 8$ roughly doubles to triples the cost of TWAP (Table 5: +3.5 and +1.7 bps on TWAP costs of 1.9 and 1.5 bps). This is the expected direction: the equilibrium of Section 3.4 describes many sellers racing each other, and a lone seller who internalises their own impact has no one to race. Injecting the equilibrium aggregate as one order prices the cost of that race. The classical single-agent, risk-neutral answer (TWAP) is what direct search recovers in every stable world at the size where the curve is resolvable.

Model structure. Two assumptions of the linear model can be checked. Cost grows less than linearly in size in three of four stable worlds (Table 8; exponents 0.5 to 0.7, with 0.96 in A6), in line with the concavity found empirically (Tóth et al., 2011); these are point estimates from three sizes. We could not estimate impact decay: five minutes after the deadline the displacement estimates are too noisy to distinguish full persistence from full reversion (Figure 4).

Buy versus sell. No stable world shows a significant difference between the cost of buying and selling at any size (Table 8).

4.6 TRADES arm

The TRADES arm is small: 6 seeds, each with an unperturbed twin and five interventions (36 episodes of about 15 minutes each). The parent order is 31,562 shares, 10% of the generated market's mean volume in the 10-minute window. Twin and intervention were identical before the

Table 8: Shape of TWAP impact cost. Size exponent: log–log slope of mean sell cost against size over the three sizes (1 = linear, 0.5 = square-root); a point estimate from three sizes, not reported for unstable worlds. Buy−sell: paired difference in cost (bps) with 95% interval.

World	Size exponent (sell)	Buy−sell, 1%	Buy−sell, 5%	Buy−sell, 20%
A1	0.61	-0.8 [-1.6, 0.0]	-0.9 [-2.1, 0.3]	0.0 [-1.2, 1.2]
A2	0.50	0.3 [-1.0, 1.5]	-0.7 [-2.0, 0.8]	-0.4 [-1.9, 1.0]
A3	–	309.8 [-73.4, 1016.6]	15.6 [-78.7, 114.0]	1564.9 [214.2, 3671.7]
A4	–	-9.6 [-28.5, 5.4]	-4.2 [-10.6, 3.1]	166.6 [-26.6, 477.5]
A5	0.66	0.0 [-0.7, 0.8]	0.2 [-0.6, 1.0]	1.1 [-0.1, 2.1]
A6	0.96	-1.2 [-4.3, 1.8]	-2.0 [-6.3, 2.0]	-1.0 [-5.5, 3.5]

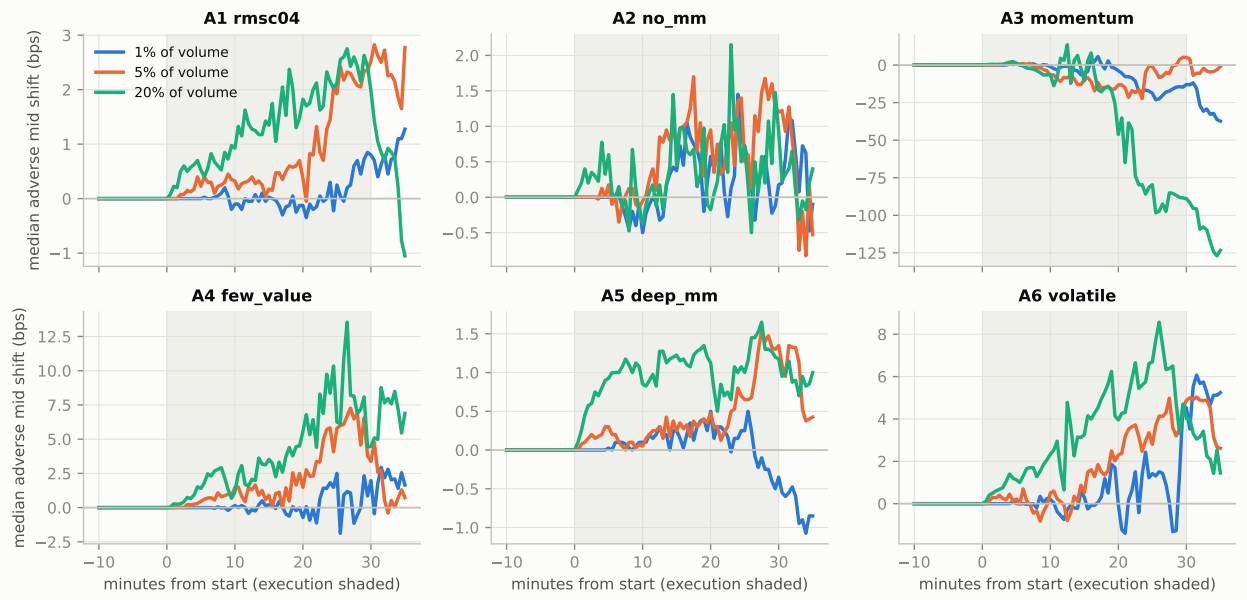


Figure 4: Median adverse displacement of the mid price relative to the twin, during and after a TWAP sell. A positive value means the price is lower than it would have been.

first child order in 100% of runs and differed afterwards in all of them, so the generated flow does respond to the injected orders.

Table 9: The 10-minute task (sell 10% of background volume in 20 slices) in every ABIDES world and in the retrained TRADES market T1. Paired impact cost in bps, mean ± 95% half-width. Bold: cheapest schedule in each column.

Policy	A1	A2	A3	A4	A5	A6	T1
Schedule $b = -2$ (back-loaded)	0.7 ±0.2	**0.4 ±0.3**	11.8 ±9.1	1.0 ±1.1	**0.5 ±0.2**	1.2 ±1.2	41.6 ±21.2
Schedule $b = 0$ (TWAP)	0.7 ±0.3	0.4 ±0.3	4.6 ±10.8	1.2 ±1.2	0.6 ±0.2	0.8 ±0.7	50.6 ±27.2
Schedule $b = 2$	**0.5 ±0.3**	0.6 ±0.3	4.4 ±6.3	**0.7 ±0.6**	0.6 ±0.1	**0.3 ±0.6**	44.2 ±28.8
Schedule $b = 8$ (front-loaded)	1.1 ±0.3	1.2 ±0.3	**3.8 ±2.3**	1.5 ±0.6	0.7 ±0.1	1.6 ±0.6	**26.6 ±16.6**
Immediate	2.1 ±0.4	1.9 ±0.5	5.3 ±2.3	2.5 ±0.9	1.1 ±0.2	4.5 ±1.4	15.7 ±2.4
Episodes per cell	40	40	40	40	40	40	6

Table 9 puts the TRADES market next to the ABIDES worlds on the identical task. The two families give opposite advice. In every stable ABIDES world the immediate order is clearly the most expensive policy and the schedules are close to one another. In the TRADES market the ordering is reversed: the immediate order and the most front-loaded schedule are the cheapest, at 15.7 bps and 26.6 bps ($b = 8$), and TWAP is the most expensive, at 50.6 bps. Paired by seed, $b = 8$ differs from TWAP by -24.1 bps (95% interval [-47.5, -6.6]; cheaper in 6 of 6 seeds), and the immediate order by -34.9 bps ([-63.5, -8.8]). The estimates are noisy: two seeds have TWAP costs near 100 bps. The generated market is also far more impact-sensitive than any ABIDES world: selling 10% of volume shifts the mid by 49 to 97 bps by the deadline depending on the schedule, against at most 0.9 bps under TWAP in the stable ABIDES worlds, and the shift under TWAP is still 76 bps two minutes later. Selecting the schedule in the TRADES market and deploying it in a stable ABIDES world costs between 0.1 and 1.3 bps of regret; selecting in a stable ABIDES world and deploying in the TRADES market costs between 15.0 and 17.7 bps. These are point estimates from 6 seeds with no train/test split.

So here is a pair of simulators that do disagree materially. The disagreement, however, has a mundane source, and it is one that realism statistics catch. The retrained generator does not hold the market it inherits. Across unperturbed twins the quoted spread is 5.6 bps in the last minute of real replay, 10.6 bps in the first generated minute, 17.1 bps after five minutes and 30.6 bps in the last minute of the task. A seller who trades early crosses a spread close to the real one; a seller who waits crosses a spread several times wider. Front-loading wins because the simulated market deteriorates with time since hand-over, not because of anything about impact. Table 10 shows the same thing in aggregate: over the task window the generated market's spread and volatility are well above those of the real day it was trained on.

Table 10: Unperturbed retrained-TRADES market against the real day it was trained on, over the task window. The real window lies in the training split.

Statistic (09:46–09:56)	TRADES mean	TRADES s.d.	Real INTC
Quoted spread (bps)	24.06	13.23	3.66
10 s return volatility (bps)	7.29	1.25	3.92
Return autocorrelation, lag 1	-0.38	0.09	-0.15
|return| autocorrelation, lag 1	0.23	0.08	-0.15
Traded volume (shares)	315618	39337	247627

We stress what this does and does not show. It is evidence about a TRADES architecture trained for 812 steps on one day, and says nothing about the published model. But it illustrates the second way, besides instability, in which a simulator can mislead an execution study: a generator whose state drifts away from its conditioning imposes a time preference of its own on every schedule evaluated inside it. A spread-versus-time plot of unperturbed runs is enough to detect it.

5 Discussion

What the evidence supports. Among the four stable ABIDES configurations we could not detect disagreement about which execution schedule is best. Rankings correlate at about 0.6 once the order is large enough to measure, there are no significant reversals, all four select TWAP at 20% of volume, and the largest transfer regret is 1.5 bps. The configurations do differ in how steeply they penalise urgency, by a factor of about four between the reference world and the world without market makers, so they would disagree about a risk-averse trader's trade-off even though they agree about a risk-neutral one. We did not run that experiment.

Where simulators did disagree. The one material disagreement in the study is between families: the retrained TRADES market favours selling at once, every stable ABIDES world favours spreading, and carrying an ABIDES-selected schedule into the TRADES market costs 15.0 to 17.7 bps on point estimates. We do not read this as a property of diffusion-based generators. The TRADES arm differs from the ABIDES arm in stock, price level, calibration and training budget at once, and the direct cause, a spread that widens with time since hand-over, is a defect of an under-trained model. What transfers is the diagnostic: a simulator can impose a time preference of its own, and it shows up in unperturbed runs.

What it does not support. The hypothesis that motivated the study, that plausible simulators recommend materially different schedules, is not supported for these configurations at this resolution. We also cannot claim they are equivalent: the intervals in Table 4 would hide differences of one or two basis points at 5% and several at 20% in A6. And six configurations of one simulator, sharing an exchange and most of their agents, are a narrow sample of "plausible simulators".

Stability before realism. The most useful practical finding is the cheapest one to check. Two configurations obtained by changing a single population parameter of the reference configuration produce markets whose unperturbed price paths are too wild for any execution experiment, and one of them contains seeds that never terminate. Neither is flagged by spread, kurtosis or volatility clustering, and both sit nearer to the real reference day than the stable worlds do. An unperturbed-stability check, and a twin-pairing noise check like Table 3, cost a few hundred episodes and should precede any policy evaluation.

On the MFG as a decision rule. Calibrating a two-parameter impact model from probe trades and reading the schedule off a closed form is attractive because it is cheap. In these simulators it is also fragile: the schedule depends on a ratio whose numerator is statistically indistinguishable from zero in most cells. Direct search over the same schedule family, with paired twins, used the same episodes and gave a stable answer.

6 Limitations

- **No real-market ground truth.** Regret is simulator-to-simulator. We cannot say which world is right, only whether they agree. The single INTC day is a reference point for stylised facts, not a validation.
- **One simulator family.** All ABIDES worlds share an exchange, an oracle and most agent code; TRADES runs inside a legacy ABIDES exchange. ABIDES-MARL is absent because we found no public code.
- **TRADES is our retraining, not the published model.** It was trained for 812 steps on one stock-day; its conclusions do not transfer to the authors' checkpoints. The TRADES and ABIDES arms differ in stock, price level, tick-to-price ratio and calibration, so differences between them cannot be attributed to the generator.
- **Statistical power.** 38 seeds, split in half for selection and evaluation. Differences below roughly 1 bp at 5% and 1 to 3 bps at 20% are not detectable in stable worlds, and nothing is detectable in unstable ones. No multiplicity correction was applied.
- **Narrow policy class.** Market orders only, a five-point schedule grid, no limit orders, no fees, no reinforcement learning, risk-neutral objective.
- **Post-hoc choices.** The stability rule was set after seeing the data, and two configurations were changed during piloting for runtime reasons (36 instead of 60 momentum traders; market makers at the default instead of a 10-second refresh interval). Runaway seeds are excluded, which conditions on an outcome.
- **MFG simplification.** We test the risk-neutral homogeneous equilibrium injected as one meta-order. Heterogeneous beliefs, latent alpha and filtering, which are the substance of Casgrain and Jaimungal (2020), are not exercised.
- **Novelty.** We ran targeted searches, not a systematic review, and make no claim to be first.

7 Conclusion

We set out to show that plausible market simulators recommend different trades. For four stable ABIDES configurations they do not, as far as 38 paired seeds can tell: they agree that an order of 20% of volume should be spread evenly, and that below that size the schedule barely matters as long as the order is not sent all at once. Two further configurations, one parameter away from the reference, are too unstable to say anything, and conventional realism statistics do not reveal it. An MFG calibrated per world does give different answers in different worlds, but the differences are calibration noise, and its front-loaded prescription is measurably worse than TWAP for a single order. A TRADES market retrained on one day of data recommends the opposite of ABIDES, for a reason that has to do with the generator drifting and not with execution. The harness, the paired-twin estimator and all results are released so the comparison can be extended to simulators we could not run.

References

Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3 (2):5–39, 2001.

Selim Amrouni, Aymeric Moulin, Jared Vann, Svitlana Vyetrenko, Tucker Balch, and Manuela Veloso. ABIDES-Gym: Gym environments for multi-agent discrete event simulation and application to financial markets. In *Proceedings of the Second ACM International Conference on AI in Finance*, 2021.

Tucker Hybinette Balch, Mahmoud Mahfouz, Joshua Lockhart, Maria Hybinette, and David Byrd. How to evaluate trading strategies: Single agent market replay or multiple agent interactive simulation? *arXiv preprint arXiv:1906.12010*, 2019.

Leonardo Berti, Bardh Prenkaj, and Paola Velardi. TRADES: Generating realistic market simulations with diffusion models. *arXiv preprint arXiv:2502.07071*, 2025.

David Byrd, Maria Hybinette, and Tucker Hybinette Balch. ABIDES: Towards high-fidelity market simulation for AI research. *arXiv preprint arXiv:1904.12066*, 2019.

Pierre Cardaliaguet and Charles-Albert Lehalle. Mean field game of controls and an application to trade crowding. *Mathematics and Financial Economics*, 12(3):335–363, 2018.

Philippe Casgrain and Sebastian Jaimungal. Mean-field games with differing beliefs for algorithmic trading. *Mathematical Finance*, 30(3):995–1034, 2020.

Patrick Cheridito, Jean-Loup Dupret, and Zhexin Wu. ABIDES-MARL: A multi-agent reinforcement learning environment for endogenous price formation and execution in a limit order book. *arXiv preprint arXiv:2511.02016*, 2025.

Andrea Coletta, Matteo Prata, Michele Conti, Emanuele Mercanti, Novella Bartolini, Aymeric Moulin, Svitlana Vyetrenko, and Tucker Balch. Towards realistic market simulations: a generative adversarial networks approach. In *Proceedings of the Second ACM International Conference on AI in Finance*, 2021.

Ruihong Huang and Tomas Polak. LOBSTER: Limit order book reconstruction system. Technical report, Humboldt Universität zu Berlin, 2011.

Xuancheng Huang, Sebastian Jaimungal, and Mojtaba Nourian. Mean-field game strategies for optimal execution. *Applied Mathematical Finance*, 26(2):153–185, 2019.

Peer Nagy, Sascha Frey, Kang Li, Bidipta Sarkar, Svitlana Vyetrenko, Stefan Zohren, Ani Calinescu, and Jakob Foerster. LOB-Bench: Benchmarking generative AI for finance – an application to limit order book data. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

Bence Tóth, Yves Lempérière, Cyril Deremble, Joachim de Lataillade, Julien Kockelkoren, and Jean-Philippe Bouchaud. Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2):021006, 2011.

Svitlana Vyetrenko, David Byrd, Nick Petosa, Mahmoud Mahfouz, Danial Dervovic, Manuela Veloso, and Tucker Balch. Get real: Realism metrics for robust limit order book market simulations. In *Proceedings of the First ACM International Conference on AI in Finance*, 2020.

Yanzhi Zhang, Yu Ma, Yilin Cheng, Jian Li, and Yitong Duan. M3: A state-event generative foundation model for market microstructure dynamics. *arXiv preprint arXiv:2608.19227*, 2026.

A Deviations from the pre-specified plan

The study followed a written research plan (`abides_marl_trades_research_plan.md` in the repository). Table 11 records where the executed study departs from it.

Table 11: Planned versus executed.

Plan	What was done, and why
Three systems: ABIDES, ABIDES-MARL, TRADES	ABIDES (six configurations) and TRADES. No public ABIDES-MARL implementation was found, so that arm is absent.
TRADES with the authors' checkpoints	The checkpoint folder has been deleted. TRADES was retrained on the one public LOBSTER sample day for 812 steps.
Go/no-go gates before experiments	Gates G1 (reproducible), G2 (interactive) and G3 (common policy) pass for all ABIDES worlds; for TRADES, twin and intervention were identical before the first order and different afterwards in every run. G4 (comparable calibration) does not hold between ABIDES and TRADES.
Realism against held-out real data	No licensed data was available. The single public INTC day is used as one reference point; the main analysis uses observational distance *between* worlds.
Sizes 0.1%, 0.5%, 1% of volume	1%, 5% and 20%. Below 1% a 60-slice schedule trades about one share per slice in these worlds.
RL policy trained per simulator	Replaced by exhaustive search over a one-parameter schedule family and by an MFG calibrated per world. No RL was trained.
Matched-state action comparison	Not done: neither system offers a state snapshot. Decisions are compared from matched initial conditions only.
Multiple configurations per system	Six for ABIDES, one for TRADES.
Pre-registration, power analysis	Not done. Seed counts were set by the compute available overnight on one laptop.
Fees, latency sensitivity	Not done. Zero fees, ABIDES default latency.
Three researchers, 9–13 weeks	One overnight session on one laptop.

B Transfer-regret matrices

Table 12: Transfer regret R_{ij} in bps, parent order 1% of background volume (test seeds). † paired 95% interval excludes zero.

Optimised in ↓ / evaluated in →	A1	A2	A3	A4	A5	A6
A1 ($b^\star = 8$)	0	-0.9	9.2	0.1	0.2	-0.7
A2 ($b^\star = -2$)	-0.4	0	-34.0	-1.7	0.0	-1.8
A3 ($b^\star = 4$)	0.0	-1.1	0	0.0	0.1	-0.8
A4 ($b^\star = 4$)	0.0	-1.1	0.0	0	0.1	-0.8
A5 ($b^\star = -2$)	-0.4	0.0	-34.0	-1.7	0	-1.8
A6 ($b^\star = 0$)	0.0	-0.5	11.4	-6.1	0.1	0

Table 13: Transfer regret R_{ij} in bps, parent order 5% of background volume (test seeds). † paired 95% interval excludes zero.

Optimised in ↓ / evaluated in →	A1	A2	A3	A4	A5	A6
A1 ($b^\star=4$)	0	0.1	-16.9	0.0	-0.4	0.0
A2 ($b^\star=-2$)	-1.0†	0	0.0	-6.2	0.0	1.5
A3 ($b^\star=-2$)	-1.0†	0.0	0	-6.2	0.0	1.5
A4 ($b^\star=4$)	0.0	0.1	-16.9	0	-0.4	0.0
A5 ($b^\star=-2$)	-1.0†	0.0	0.0	-6.2	0	1.5
A6 ($b^\star=4$)	0.0	0.1	-16.9	0.0	-0.4	0

Table 14: Transfer regret R_{ij} in bps, parent order 20% of background volume (test seeds). † paired 95% interval excludes zero.

Optimised in ↓ / evaluated in →	A1	A2	A3	A4	A5	A6
A1 ($b^\star=0$)	0	0.0	0.0	-37.5	0.0	0.0
A2 ($b^\star=0$)	0.0	0	0.0	-37.5	0.0	0.0
A3 ($b^\star=0$)	0.0	0.0	0	-37.5	0.0	0.0
A4 ($b^\star=4$)	2.1	-0.5	1816.9	0	1.2†	0.3
A5 ($b^\star=0$)	0.0	0.0	0.0	-37.5	0	0.0
A6 ($b^\star=0$)	0.0	0.0	0.0	-37.5	0.0	0

C Implementation notes

ABIDES. We use `abides-core` and `abides-markets` from the `abides-jpmc-public` repository. Its order-size model depends on an old release of the `pomegranate` library that no longer installs; we substitute a small module that samples the same mixture distribution. The world builder mirrors the seeding order of the RMSC04 configuration and appends the execution agent last.

Stalls and runaways. Each episode has a wall-clock limit and is retried once. In total, 1 unperturbed and 5 intervention episodes hit the wall-clock limit on every attempt (A3 seed 1, A3 seed 118, A3 seed 16); they are excluded together with all other episodes of the same seed, leaving 38 of 40 seeds in the main analysis.

DeepMarket. The patch shipped with the code (i) removes non-blocking device transfers that corrupt integer tensors on Apple GPUs, (ii) restores the aggressor's fill notification for execution agents, (iii) shrinks a pre-allocated order-identifier pool, and (iv) allows loading our own checkpoint. The stock data builder splits by whole days and cannot process a one-day dataset; our training script splits the day chronologically instead. The simulation normalises inputs with whole-day statistics, whereas training used statistics of the first 85% of the day; we did not correct this small mismatch.

A sign error in the pre-existing MFG solver. The repository this study extends contains an iterative forward–backward solver for the MFG. With selling rate $\nu = -\dot{q}$, its backward step implements $\dot{\nu} = (2\phi q - A - \lambda\bar{\nu})/(2a)$, whereas the Euler–Lagrange condition gives $\dot{\nu} = (A - \lambda\bar{\nu} - 2\phi q)/(2a)$: the inventory-penalty and drift terms have the wrong sign. Numerically, a more risk-averse seller trades *later* under that solver and sells sooner when the price drifts up. We did not modify it; this paper uses the closed form of Section 3.4.